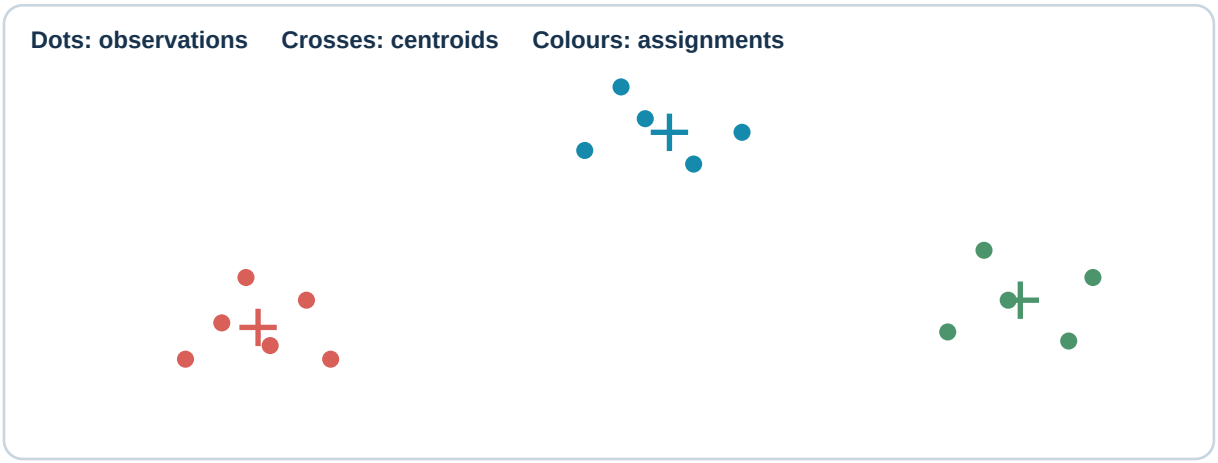
FROM CLUSTERS TO PROBABILISTIC MODELS

K-Means, Gaussian Mixture Models and the Expectation–Maximization Algorithm



Derivation-first study notes with intuition, diagrams, worked examples, applications, limitations and common misconceptions.

Learning path

1. K-means clustering in detail
2. Why k-means is not always enough
3. Gaussian Mixture Models (GMMs)
4. Expectation–Maximization (EM)
5. General EM identity, convergence and exam revision

How to use these notes

Follow the chapters in order. The boxed equations are the key results; the derivations explain why they hold. The material incorporates the supplied screenshots on k-means, the alternative view of EM, the general latent-variable formulation, and the k-means/GMM relationship.

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2. K-means objective and derivations

3. Convergence, initialization and choosing K

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1. Clustering and its purpose

Clustering is an unsupervised-learning task: observations are available but target labels are not. The goal is to discover groups whose members are similar under a chosen representation and distance or probability model.

```
X = {x_1, ..., x_N}, x_n in R^D
```

Notation: N is the number of observations, D the number of features, and K the number of clusters or mixture components. Typical uses include customer segmentation, document clustering using word-frequency vectors, image segmentation, exploratory analysis and prototype-based compression.

A good clustering usually has high within-cluster similarity and low between-cluster similarity. This is not an absolute guarantee of meaningful real-world categories: cluster IDs need interpretation.

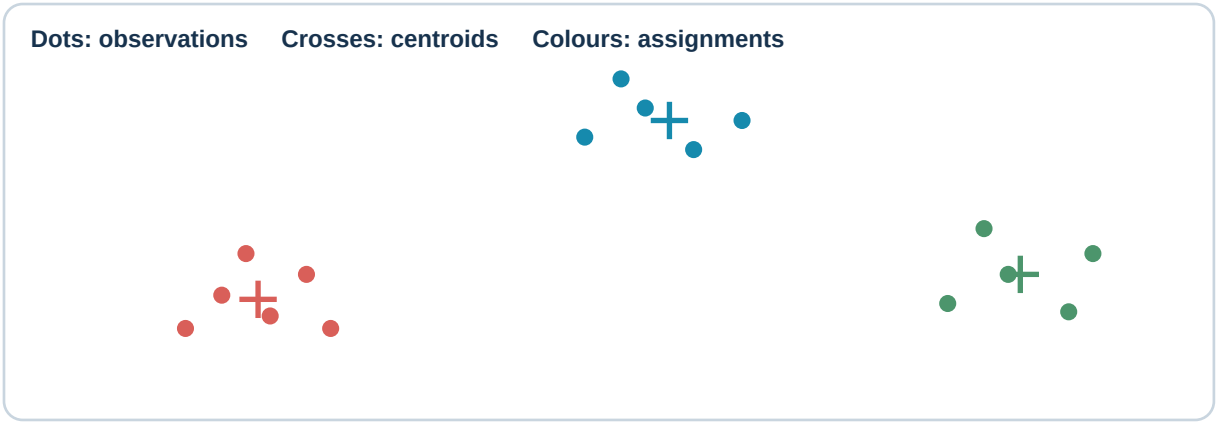


Figure 1. Schematic clusters: dots are observations, crosses are centroids and colours indicate assignments.

2. K-means: objective and derivations

2.1 Objective and indicators

K-means learns centroids mu_1,...,mu_K and an assignment c_n in {1,...,K} for each observation. Its distortion, also called within-cluster sum of squares, is

```
J(c, mu) = sum_{n=1}^N ||x_n - mu_{c_n}||^2
```

Use binary indicators r_nk=1 if point n belongs to cluster k, otherwise 0. Exactly one indicator is one for each observation:

```
r_nk in {0,1}, sum_{k=1}^K r_nk = 1
```

```
J(R, mu) = sum_n sum_k r_nk ||x_n - mu_k||^2
```

The indicator formulation works because all unassigned-cluster terms are multiplied by zero.

2.2 Assignment step

Fix the centroids. Each point's contribution is minimised by choosing the cluster with the smallest squared distance. Thus

```
c_n = arg min_k ||x_n - mu_k||^2
```

This is an exact minimiser over assignments for fixed centroids. Squared distance and distance give the same nearest-centroid choice because squaring is monotonic for nonnegative distances.

2.3 Voronoi geometry

For two centroids mu_a and mu_b, the equal-distance boundary satisfies ||x-mu_a||^2=||x-mu_b||^2. Expand both sides and cancel x^T x:

```
2 x^T (mu_b - mu_a) = ||mu_b||^2 - ||mu_a||^2
```

This is linear in x: a line in 2-D, a plane in 3-D, or a hyperplane in D dimensions. Nearest-centroid regions are Voronoi cells, helping explain why k-means struggles with curved, non-convex groups.

2.4 Derive the centroid update

Hold assignments fixed. For cluster k, let J_k=sum_{n:c_n=k} ||x_n-mu_k||^2. Since the vector derivative of ||x-mu||^2 with respect to mu is 2(mu-x),

```
grad_{mu_k} J_k = 2 sum_{n:c_n=k} (mu_k - x_n)
```

Set the gradient to zero and let N_k be the number of points in cluster k:

```
N_k mu_k - sum_{n:c_n=k} x_n = 0
```

```
mu_k = (1/N_k) sum_{n:c_n=k} x_n = (sum_n r_nk x_n)/(sum_n r_nk)
```

Why is this a minimum? The Hessian is 2 N_k I_D, positive definite for a nonempty cluster. Thus the mean is the unique minimiser for fixed assignments. The arithmetic mean minimises squared Euclidean distances; a sum of unsquared distances generally leads to a geometric median.

3. Convergence, initialization and choosing K

3.1 Why distortion never increases

The assignment step minimises J for fixed centroids. The update step minimises J for fixed assignments. Hence

```
J(c^(t+1),mu^(t)) <= J(c^(t),mu^(t))
```

```
J(c^(t+1),mu^(t+1)) <= J(c^(t+1),mu^(t))
```

```
J^(t+1) <= J^(t), J^(t) >= 0
```

The objective values converge because they are non-increasing and bounded below. This does not guarantee a global minimum: the joint problem is non-convex. With finite data and consistent tie/empty-cluster handling, assignments typically stabilise.

3.2 Initialization

Poor initial centres can place multiple centroids in one natural group and leave another group poorly represented. Use k-means++ or multiple random restarts and choose the run with the lowest final distortion. Multiple restarts improve the chance of a good solution but do not prove global optimality.

```
Best restart = arg min_r J_r
```

3.3 Choosing K: elbow and silhouette

Globally optimal distortion cannot increase with K: an extra cluster can reproduce an old solution by remaining unused. At K=N, distortion can be zero, but this over-fragments the data. The elbow method seeks the bend where extra clusters produce diminishing returns; the bend may not be clear.

```
s(i) = [b(i) - a(i)] / max(a(i), b(i))
```

Here a(i) is average distance to points in the same cluster, and b(i) is the smallest average distance to another cluster. Silhouette values near +1 indicate separation, around 0 a boundary point, and below 0 a possible misassignment.

3.4 Empty clusters

If no point is assigned to a centroid, the mean update has a zero denominator. Implementations commonly reinitialise it or apply another explicit empty-cluster policy.

4. Why GMM is useful

K-means optimises distances to centres. It does not explicitly model cluster orientation, covariance, or membership uncertainty. A long tilted cloud is not described by a centre alone; overlapping groups are forced into hard assignments.

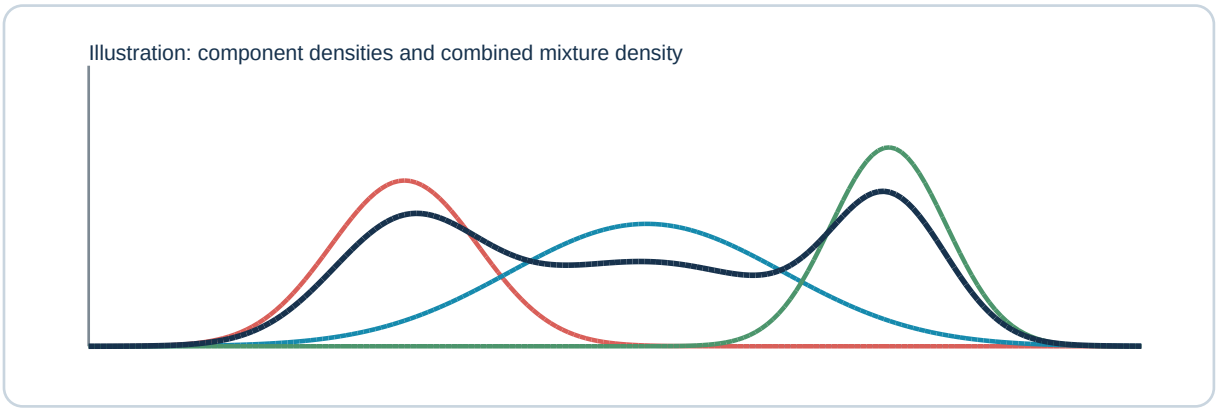


Figure 2. Component densities combine into a multi-peaked mixture density.

Use k-means when compact groups, fast hard partitions and Euclidean distance are suitable. **Consider GMM** when soft membership, uncertainty, tilted ellipses or probabilistic density estimates are useful.

A GMM is not a universal fix: finite Gaussian mixtures can fit poorly, EM can find local optima, and covariance estimates can become degenerate.

5. Gaussian Mixture Models

5.1 Generative story

For each observation, first sample a hidden component label z_n from a categorical distribution with probabilities pi_1,...,pi_K, then sample x_n from that component's Gaussian.

$$p(z_n=k)=\pi_k,\ \pi_k \geq 0,\ \sum_k \pi_k = 1$$

$$p(x_n \mid z_n=k) = \mathcal{N}(x_n \mid \mu_k, \Sigma_k)$$

5.2 Mixture density and Gaussian

$$p(x) = \sum_{k=1}^{K} \pi_k \mathcal{N}(x \mid \mu_k, \Sigma_k)$$

$$\mathcal{N}(x|\mu,\Sigma) = (2\pi)^{-D/2} |\Sigma|^{-1/2} \exp\left[-\frac{1}{2}(x-\mu)^T \Sigma^{-1}(x-\mu)\right]$$

The mean locates a component; covariance controls its spread and orientation. Diagonal covariance models axis-aligned variances; full covariance can model correlations and tilted ellipses.

5.3 Likelihood

$$\ell(\theta) = \sum_n \log \left\{ \sum_k \pi_k \mathcal{N}(x_n \mid \mu_k, \Sigma_k) \right\}$$

The log of a sum makes direct maximisation difficult. This motivates EM, which exploits the latent component labels.

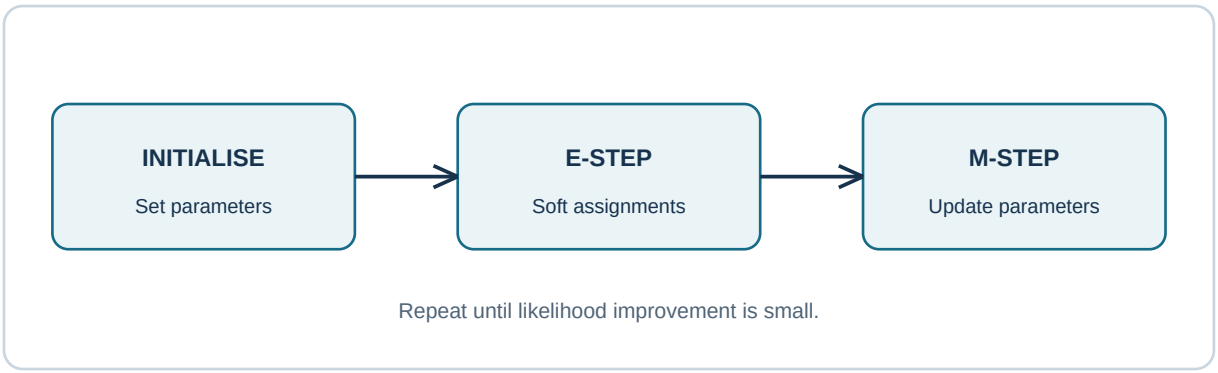
5.4 Responsibilities from Bayes' rule

$$\gamma_{nk} = p(z_n=k \mid x_n,\theta)$$

$$\gamma_{nk} = \frac{\pi_k \mathcal{N}(x_n|\mu_k,\Sigma_k)}{\sum_j \pi_j \mathcal{N}(x_n|\mu_j,\Sigma_j)}$$

For each point, responsibilities lie in [0,1] and sum to one over components. pi_k is a prior mixture weight for the population; gamma_nk is the posterior membership probability for a specific observation.

6. EM for GMM: E-step and M-step



6.1 Initialise

Choose K, means, covariance matrices and mixing weights. K-means can seed the means; multiple EM restarts are advisable because initialisation matters.

6.2 E-step

```
gamma_nk = [pi_k N(x_n|mu_k,Sigma_k)] / [sum_j pi_j N(x_n|mu_j,Sigma_j)]
```

Compute each observation's responsibility using current parameters. This is Bayes' rule: prior times density, normalised across components.

6.3 Effective counts and M-step

```
N_k = sum_{n=1}^N gamma_nk
mu_k(new) = (1/N_k) sum_n gamma_nk x_n
Sigma_k(new) = (1/N_k) sum_n gamma_nk (x_n-mu_k(new))(x_n-mu_k(new))^T
pi_k(new) = N_k/N
```

N_k is a soft count, not necessarily an integer. The means and covariances are weighted by responsibility. Since sum_k gamma_nk=1, sum_k N_k=N, so the updated mixing weights sum to one.

6.4 Repeat and numerical stability

Repeat E- and M-steps until likelihood improvement or parameter changes are small, or a maximum iteration count is reached. Use log-sum-exp for stable mixture probabilities and consider covariance regularisation Sigma_k + epsilon I when components become nearly singular.

7. General EM and the lower-bound identity

7.1 Observed and hidden variables

Let X be observed data, Z latent variables and theta parameters. The observed likelihood marginalises the latent variables:

```
p(X|theta) = sum_Z p(X,Z|theta)
```

If Z were known, the complete-data log-likelihood log p(X,Z|theta) may be easier to optimise. EM uses its expectation under the current posterior.

7.2 Q function and algorithm

```
Q(theta,theta_old) = E_{Z|X,theta_old}[ log p(X,Z|theta) ]
```

```
theta_new = arg max_theta Q(theta,theta_old)
```

E-step: compute p(Z|X,theta_old) and the expectation. M-step: maximise Q over candidate theta. The E-step does not generally replace Z with just its most likely value; it averages over latent possibilities.

7.3 Derive the identity from the supplied screenshot

For any distribution q(Z) with appropriate support, define

```
L(q,theta) = sum_Z q(Z) log [p(X,Z|theta)/q(Z)]
```

```
KL(q || p) = sum_Z q(Z) log [q(Z)/p(Z|X,theta)]
```

Bayes' rule gives log p(Z|X,theta)=log p(X,Z|theta)-log p(X|theta). Rearranging, adding and subtracting log q(Z), and taking expectation under q gives

```
log p(X|theta) = L(q,theta) + KL(q || p(Z|X,theta))
```

KL divergence is nonnegative, so L is a lower bound on the observed log-likelihood. The bound is tight exactly when q(Z)=p(Z|X,theta), subject to the usual support conditions.

7.4 Why the E-step makes the bound tight

At the current parameters, choose q(Z)=p(Z|X,theta_old). The KL term is then zero, so the bound equals the current log-likelihood. In the M-step, maximise the bound while q is fixed. The next E-step resets the bound so that it touches the likelihood at the new parameters.

8. Why EM improves likelihood

Let q_old(Z)=p(Z|X,theta_old). At the old parameters, the bound is tight. An exact M-step maximises the bound and therefore cannot lower it:

```
log p(X|theta_new) >= L(q_old,theta_new) >= L(q_old,theta_old) = log p(X|theta_old)
```

Thus standard exact EM does not decrease observed-data log-likelihood. Generalised EM also preserves monotonicity when its M-step increases Q. But monotonicity is not global optimality: EM may reach a local maximum or stationary point. In unconstrained Gaussian mixtures, likelihood can be unbounded when a covariance collapses around a data point.

Practical safeguards: multiple initialisations, monitor likelihood, inspect tiny components, regularise covariance matrices, and use sensible model selection. EM can be slower than direct gradient optimisation near a solution.

9. K-means as a limiting case of GMM

Suppose each GMM component has shared spherical covariance Sigma_k=epsilon I. Then

```
gamma_nk = [pi_k exp(-||x_n-mu_k||^2/(2 epsilon))] / [sum_j pi_j exp(-||x_n-mu_j||^2/(2 epsilon))]
```

As epsilon approaches zero, the exponential for the nearest mean decays slowest. If the nearest centroid is unique and relevant weights are nonzero, its responsibility tends to one and the others to zero:

```
epsilon -> 0: gamma_nk -> r_nk in {0,1}
```

Soft assignments become hard, responsibility-weighted means become ordinary means, and mixing weights become cluster proportions. The expected complete-data log-likelihood approaches the k-means distortion objective up to constants and scale. This is a limiting relationship under restrictive assumptions, not equivalence between arbitrary GMM and k-means.

Quick comparison

Property	K-means	GMM + EM
Membership	Hard 0/1	Soft responsibilities
Shape	Voronoi cells	Gaussian ellipses
Objective	Minimise squared distances	Maximise likelihood
Spread	Not explicitly modelled	Covariance estimated
Centre update	Arithmetic mean	Weighted mean

10. Worked example, applications and misconceptions

10.1 Tiny k-means example

Data X={1,2,3,8,9,10}, K=2. Start with mu_1=2 and mu_2=9. The assignments are C_1={1,2,3}, C_2={8,9,10}. Updated means remain 2 and 9.

```
J = (1-2)^2 + (2-2)^2 + (3-2)^2 + (8-9)^2 + (9-9)^2 + (10-9)^2 = 4
```

Since assignments and centres remain unchanged, this initialisation has converged.

10.2 Common misconceptions

1. EM always finds the global maximum. False: it improves likelihood monotonically under standard exact steps but may find a local solution.

2. E-step means taking an expectation of parameters. False: it computes the posterior over latent variables and forms an expectation of the complete-data log-likelihood.

3. Responsibilities equal mixing coefficients. False: pi_k is a population-level prior weight; gamma_nk is point-specific posterior membership.

4. GMM components are real-world classes. Not necessarily. A class may need multiple components, or components may not align with semantic labels.

5. More clusters always mean a better model. Training likelihood may improve with flexibility, but extra components can overfit or become degenerate.

6. GMM fixes all shape problems. False: it models Gaussian components; it can still fit poorly or converge to problematic solutions.

10.3 Applications

K-means: customer segmentation, document grouping, image segmentation and prototypes. GMM/EM: soft clustering, density estimation, probabilistic segmentation, anomaly scoring and latent-variable or missing-data estimation. Low density is a reason to investigate, not proof that a point is erroneous.

11. Formula sheet and self-test

Key formulas

```
K-means J = sum_n sum_k r_nk ||x_n-mu_k||^2
Assignment: c_n = arg min_k ||x_n-mu_k||^2
K-means mean: mu_k = sum_n r_nk x_n / sum_n r_nk
GMM density: p(x) = sum_k pi_k N(x|mu_k,Sigma_k)
Responsibility: gamma_nk = pi_k N(x_n|mu_k,Sigma_k) / sum_j pi_j N(x_n|mu_j,Sigma_j)
N_k = sum_n gamma_nk; mu_k = sum_n gamma_nk x_n / N_k
Sigma_k = sum_n gamma_nk (x_n-mu_k)(x_n-mu_k)^T / N_k
pi_k = N_k/N
Q(theta,theta_old) = E_{Z|X,theta_old}[log p(X,Z|theta)]
log p(X|theta) = L(q,theta) + KL(q || p(Z|X,theta))
```

Self-test

1. Why does differentiating the k-means objective give the arithmetic mean? 2. Prove each k-means step cannot increase distortion. 3. Why is global optimality not guaranteed? 4. Explain the elbow method. 5. Derive GMM responsibilities using Bayes' rule. 6. Explain each M-step update. 7. Derive the general EM lower-bound identity. 8. Why does exact EM not decrease likelihood? 9. Under what assumptions does GMM approach k-means? 10. Distinguish pi_k, gamma_nk and N_k.

Short answer prompts

1. Gradient 2 sum(mu_k-x_n)=0 gives the mean; Hessian 2N_k I is positive definite. 2. Each step minimises the objective over one block of variables. 3. The objective is non-convex. 4. Look for diminishing returns as K grows. 5. Posterior equals prior times likelihood, normalised. 6. Weighted mean, weighted covariance, effective proportion. 7. Apply Bayes' rule, add/subtract log q and take expectation. 8. The bound is tight at the old parameters and is increased by the M-step. 9. Shared spherical covariance tending to zero and unique nearest centroids. 10. Prior mixture weight, point-specific posterior, total effective count.

12. References and further reading

1. Requested EM explainer:
https://medium.com/@chloebee/the-em-algorithm-explained-52182dbb19d9

2. Supplied screenshots: Sections 9.3, 9.3.2 and 9.4 covering the alternative view of EM, its relationship to k-means, the general latent-variable formulation, and the likelihood/lower-bound decomposition.

3. Dempster, A. P., Laird, N. M., and Rubin, D. B. (1977). Maximum Likelihood from Incomplete Data via the EM Algorithm. Journal of the Royal Statistical Society: Series B, 39(1), 1–38.

4. Bishop, C. M. (2006). Pattern Recognition and Machine Learning. Mixture models and EM chapters.

5. Further explanation: https://quantmemo.com/concepts/em-algorithm.

These notes synthesise the provided screenshots with standard mathematical derivations. The requested Medium link is included; the document explains the concepts rather than reproducing long passages from the source.

Final memory hook: K-means alternates between nearest-centroid assignment and mean update. GMM replaces hard membership with posterior responsibilities and models covariance. EM alternates between inferring hidden variables and maximising the expected complete-data log-likelihood.